

MUHAMMAD ANAS SHEIKH

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SUMMARY

Software Engineer with 1.5+ years of experience specializing in Computer Vision and Applied Machine Learning, building and deploying real-time AI systems using CNNs, Transformers, and NVIDIA edge devices. Experienced in end-to-end ML pipelines, including data preprocessing, training, evaluation, inference, and post-processing, as well as MLOps tooling such as Docker, MLflow, and GitHub Actions (CI/CD). Currently employed at Lambda Theta (ODC), supporting enterprise AI initiatives for Elm (KSA), including the development and integration of Nuha LLM-based solutions for production-grade NLP and automation use cases.

PROFESSIONAL WORK EXPERIENCE

LAMBDA THETA | Elm – KSA (ODC)

July 2024 – present

Software Engineer

- Design and deploy AI-driven solutions using Convolutional Neural Networks (CNNs), Transformers, Mediapipe and YOLO for real-time computer vision tasks including object detection, classification, and scene analysis.
- Build and optimize scalable data extraction tools leveraging Python and large language models (LLMs) to automate web scraping and content parsing across varied domains.
- Integrate LLM and custom NLP pipelines to analyze unstructured text data, enabling contextual question answering and semantic search capabilities.
- Develop modular ML pipelines for preprocessing, inference, and postprocessing, facilitating rapid experimentation and performance benchmarking across different model architectures.
- Engineering proof-of-concept prototypes for AI applications in safety, automation, and research intelligence, accelerating idea-to-deployment workflows.
- Applying best practices in API integration, data handling, and model evaluation to ensure robustness, scalability, and maintainability of machine learning systems.

Design Intern

- Working in the Design Department, acquiring hands-on experience in Python, Machine Learning, Artificial Intelligence and Deep Learning.
- Developing expertise in Computer Vision exploring a variety of pre-trained models.
- Testing custom-built models, preparing datasets and training classification models with unique datasets, followed by fine-tuning for optimal performance.
- Actively contributing to team projects, enhancing collaborative skills and technical knowledge.
- Delivered comprehensive reports evaluating performance metrics across multiple models, providing actionable insights to drive data-informed decision-making. Enhanced model efficiency through systematic analysis and reporting.

CODSOFT

Sept 2023 – Oct 2023

Internship | Remote

- Gained hands-on experience in Java programming during an internship, developing a game application with features like random number generation, user interaction, feedback mechanism, attempt limitation, and score tracking.
- Acquired practical programming skills by developing an Automated Teller Machine simulation application. Designed an intuitive user interface for CRUD operations like withdrawals, deposits, and balance checks, integrated with validation mechanisms to ensure secure and accurate transactions.

EDUCATION

DHA SUFFA University – BSc Computer Science – GPA: 3.29

Nov 2020 – July 2024

DEVELOPMENT PROJECTS

Built deep learning systems for object detection and classification using CNNs, Transformers, and PyTorch. Optimized vision models (e.g., EfficientViT, RT-DETR, YOLO) and developed LLM-based tools for web scraping and research analysis via OpenAI. Managed ML pipelines with Docker, MLflow, and GitHub Actions, deploying real-time models on NVIDIA Jetson devices for edge AI use cases.

A full list of my projects, along with objectives, tech stacks, and outcomes, is available here:

 [View Project Portfolio](#)

- **Object Detection**

Applied data augmentation and trained a custom Ultralytics YOLO model, integrating the optimized detection pipeline into a larger AI system to enhance overall accuracy and performance.

- **Person Classifier**

Developed and fine-tuned an Ultralytics-based computer vision classifier, integrating it into a custom AI system deployed in a public environment for accurate differentiation and classification.

- **Accident Detection**

Developed a dashcam-based accident detection PoC using a pretrained MoViNet CNN, deployed it on NVIDIA Jetson Orin Nano for real-time inference, and integrated cloud-based alerting for incident notifications.

- **Weapons Detection System**

Designed and implemented a modular, multi-camera dual-model inference pipeline (YOLOWeaponDetector + YOLOPoseEstimator) with a custom EfficientViT-backed YOLOv8, rule-based alerting, evidence persistence, and API/WebSocket exposure, optimizing precision/recall performance while reducing resource utilization compared to RT-DETR.

- **Dental Classification System**

Developed and deployed a containerized Streamlit application integrating a custom-trained Ultralytics YOLO model for real-time dental image inference with configurable thresholds and automated bounding box annotations.

TECHNICAL SKILLS

- **Languages:** Python, Java, C++, C, MySQL, HTML/CSS
- **Tools:** VS Code, NetBeans, Code Blocks, GitHub, GitHub Actions, Postman, Docker, MLFlow, MySQL, Nvidia Jetson, OpenAI playground, Cap-Cut, Make.com, Notion, N8N
- **Certifications:**
 - [Supervised Machine Learning: Regression and Classification](#) (Coursera: Stanford – DeepLearning.AI)
 - [AI Agents Intensive Course with Google](#) (Kaggle - Google)
 - [Fundamentals of MLOps](#) (Kodekloud)
 - [Computer Architecture](#) (Kodekloud)

EXTRA CURRICULAR

DSU SPECTRUM 23 – Speed Debugging

- Collaborated in a team of 3 to resolve multiple complex programming issues under time constraints.
- Secured 3rd place among participating teams in the competitive challenge.

DSU SPORTS WEEK 23 – Inter-Department Cricket Tournament

- Played a key role in the Team as the Batsman, winning the tournament against various department teams.
- Lifted the winner's trophy in the event.